BRAC UNIVERSITY

Department of Computer Science and Engineering
CSE 341: Microprocessors

Examination: Quiz 2
Duration: 45 Minutes

Semester: Summer 2026
Full Marks: 20

Set A

Name:	ID:	Section:

Question 1 [7 marks]

(i) MOV AX, 0F0FH (ii) MOV BX, F0F0H (iii) ADD AX,BX

a. Deduce the values of the following status flags after the execution of the given instructions. Answer on the question paper. [6]

OF =	PF =	AF =	SF =	ZF =	CF =

b. Explain how the flag register can help a programmer detect errors or exceptional conditions during program execution in the 8086 microprocessor. [1]

Question 2 [6 marks]

a. Illustrate using a block diagram, the internal architecture of Intel 8086 and label the individual components. [4]

b. Assume you have a Microprocessor named "X", which is similar to an Intel 8086 in every way, except that the size of the Prefetch queue is 20 bytes. Now, deduce and explain the minimum number of complete instructions that can completely fill up the prefetch queue of the Microprocessor "X". Justify your answer. [1]

c. If the value of the segment registers are: DS = 3303h, SS = 7320h, CS = 5501h, ES = 1010h, then determine from which of the following addresses the instruction queue could have fetched instruction bytes. Answer on the question paper. [1]

Address	[Y/N]
62022h	
18182h	
5811Fh	
34452h	
71A82h	

Question 3 [7 marks]

Answer on the question paper.

a. Deduce which category of addressing modes the given instructions fall into. [0.5x6=3]

Instruction	Addressing Mode
i) MOV [BP+SI+2010h], AX	
ii) MOV 34H, CL	
iii) MOV CL, [BX]	
iv) MOV BL, [2000H]	
v) MOV CL, [BX+SI]	
vi) RET [1234H]	

b. Deduce the machine readable equivalent for the instruction-(i) from question 3(a). Assume, opcode for MOV = 100010. [3]

Byte 1 Byte 2

Byte 3 Byte 4

c. Deduce the corresponding hexadecimal representation of your answer obtained from 3(b). [0.5]

d. Fill the following table to indicate which byte is stored in which memory location of the RAM. Leave the unused memory spaces empty. [0.5]

Logical Address	Data
5501: 0000	
5501: 0001	
5501: 0002	
5501: 0003	
5501: 0004	
5501: 0005	

NOTE: Opcode table is given on the next page

MOD / RM	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] + d16	AH	SP
101	[DI]	[DI] + d8	[DI] + d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] + d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

Set A

1. A. OF = 0, PF = 1, AF = 0, SF = 1, ZF = 0, CF = 0
2. B.Since maximum instruction length of an instruction in Intel 8086 is 6 bytes, so 20bytes/6bytes = 3 complete instructions(each of 6 bytes) + 1 complete instruction(of 2 bytes) = 4 complete instructions.
   C. 62022h and 5811Fh = Y(since they are within the range CS:0000 to CS:FFFF), rest = N.
3. A. ii and vi = Invalid, rest check slide
   B. Byte 1: 10001001, Byte 2: 10000010, Byte 3: 00010000, Byte 4: 00100000
   C. 89 82 10 20h
   D. 5501:0000 = 89, :0001 = 82, :0002 = 10, :0003 = 20, rest = empty

Set B

1. A. OF = 0, PF = 1, AF = 1, SF = 0, ZF = 1, CF = 1
2. B. Since the maximum instruction length of an instruction in Intel 8086 is 6 bytes, so 24bytes/6bytes = 2 complete instructions(each of 6 bytes) + 1 complete instruction(of 2 bytes) = 3 complete instructions.
   C. 62022h and 71A82h = Y(since they are within the range CS:0000 to CS:FFFF), rest = N.
3. A. iv and v = Invalid, rest check slide
   B. Byte 1: 10001001, Byte 2: 10000010, Byte 3: 01100000, Byte 4: 00010000
   C. 89 82 60 10h
   D. 6201:0000 = 89, :0001 = 82, :0002 = 60, :0003 = 10, rest = empty